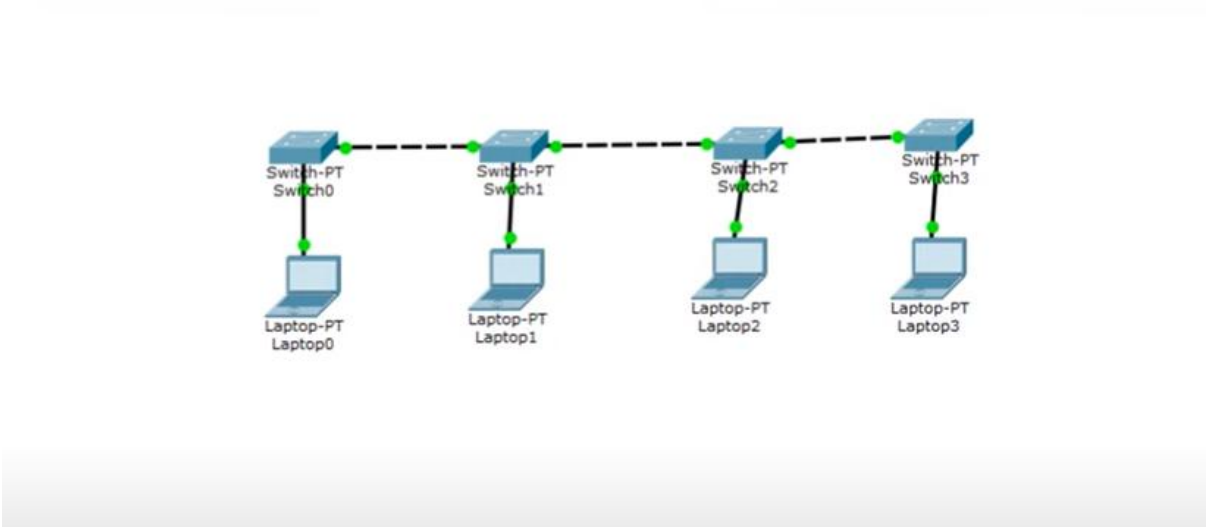
Network Topologies

1. Bus Topology Diagram

Bus Network Topology



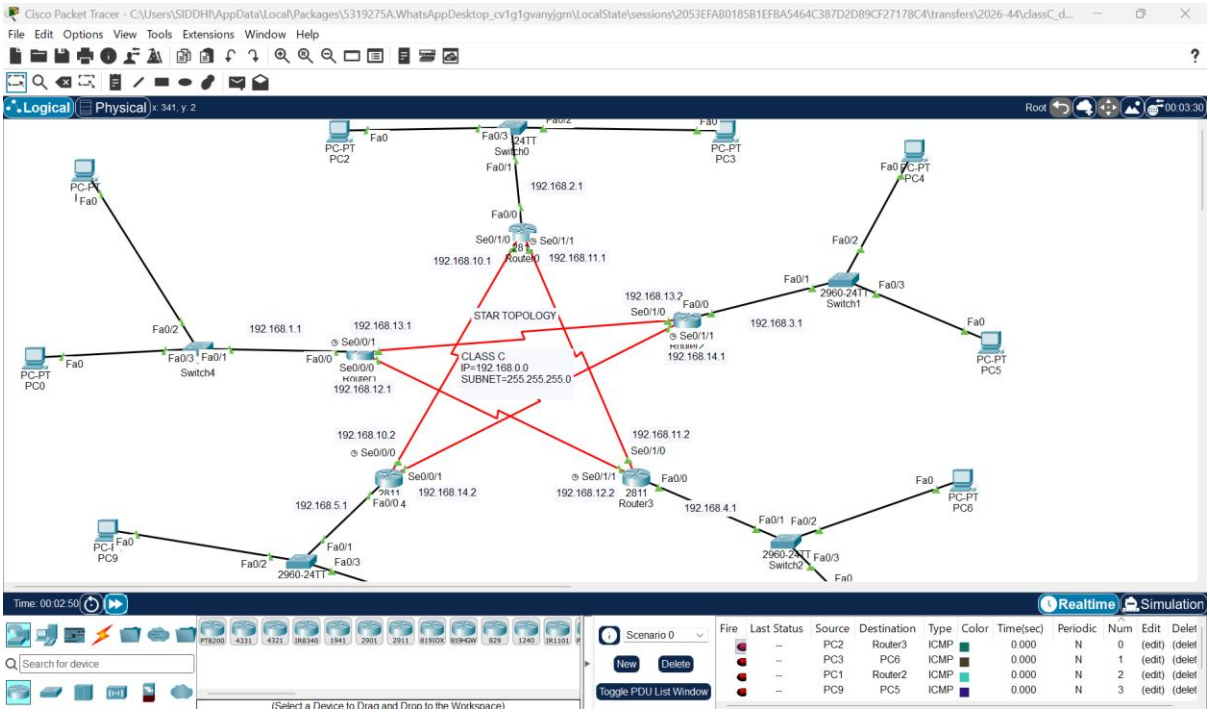
Key components: computers, backbone cable, connecting cables and terminators.

Real-world example: An older computer laboratory might use a bus topology to connect several computers through a shared coaxial cable. It uses less cable, but a backbone failure can disrupt the whole network.

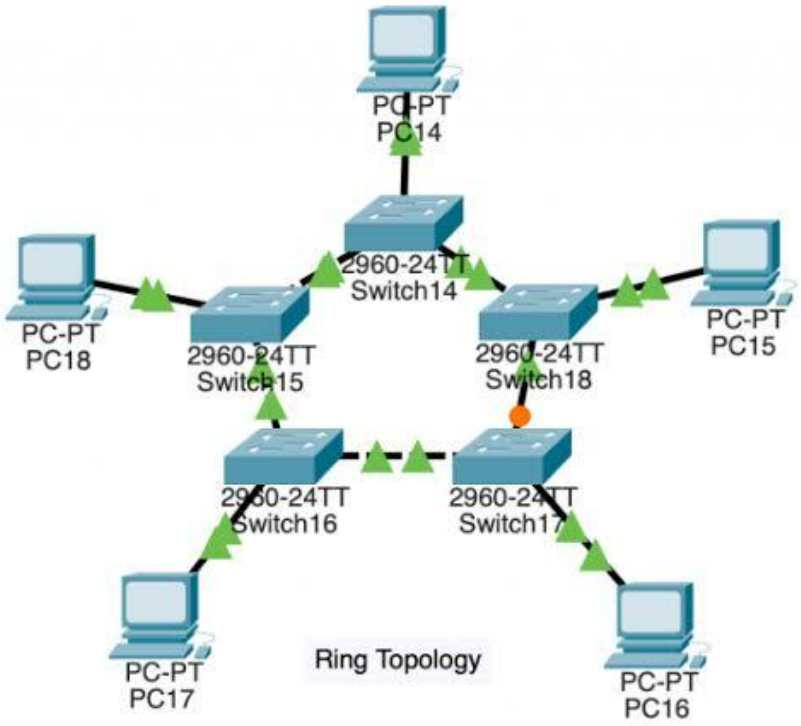
2. Create a Star Topology in Cisco Packet Tracer

Required devices: 1 switch and 4 PCs.

Star Topology Layout



3. Ring Topology Diagram



4. Mesh vs. Star Topology

Topology	Advantages	Disadvantages
Star	1. Easy to install and troubleshoot. 2. Failure of one device or cable usually does not affect the others.	1. Failure of the central switch affects the connected network. 2. Requires more cables than a bus topology.
Mesh	1. Multiple paths improve reliability. 2. Link failures can be handled by using alternative paths.	1. Expensive due to many links and ports. 2. Complex to install and maintain.

Conclusion: Star topology is a practical choice for homes, colleges and offices. Mesh topology is better suited to situations where high reliability and alternative communication paths are important.